INCH-POUND

A-A-54929
17 December 1993

COMMERCIAL ITEM DESCRIPTION

PULP PROTECTOR, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers dental pulp protectors furnished in the types specified below.

The pulp protector shall be a hard-setting, dental pulp protector suitable for use as a rigid base under restorative materials or as a direct or indirect capping agent with the following salient characteristics.

Salient characteristics:

Type:

I. Pulp Protector, Dental, Calcium Hydroxide Paste, with instrument.

II. Pulp Protector, Dental, Calcium Hydroxide Paste, without instrument.

The pulp protector shall consist of a base paste in a collapsible tube, a catalyst paste in a second collapsible tube, and a mixing pad. For type I a mixing and placement instrument shall be furnished. Type II shall be supplied without a placement instrument. The base paste, when mixed with catalyst paste shall produce a hard-setting, self-setting, non-toxic, calcium hydroxide base suitable for use as a direct or indirect pulp capping agent or subrestorative base.

When the catalyst paste and base paste are combined, the resulting paste shall be sufficient to yield 20 to 25 grams of self-setting mixed paste.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

The base paste reactive agent shall be calcium hydroxide and the catalyst paste reactive agent shall be a resin salicylate. As an alternate, the base paste reactive agent shall be a resin salicylate and the catalyst paste reactive agent shall be calcium hydroxide. The calcium hydroxide shall be in stoichiometric excess over the amount required to react with the resin salicylate.

The pulp protector shall be radiopaque to permit differentiation from dentin or caries. The color shall be dentin-like to enhance the esthetics of translucent restoratives. The pulp protector shall be capable of stimulating the formation of secondary dentin, protecting the dental pulp in the acid etch technique, structurally withstanding the forces of occlusion, and insulating the dental pulp from thermal and electrical stimuli. The pulp protector shall adhere tenaciously to the dentin surface without flaking, peeling, or dislodgement after trimming or shaping with dental instruments. The pulp protector shall be compatible with all restorative materials, and shall not interfere with the polymerization of composites or the setting reaction of luting agents, or affect the color stability or physical properties of these restorative materials. The pulp protector shall be classified as an acceptable preparation by the Council on Dental Materials and Devices of the American Dental Association.

Physical properties. The catalyst and base shall be easily mixed within 15 seconds by hand spatulation. The bench working time shall be 2 to 3 minutes at 30°C and 55% relative humidity. The pulp protector shall demonstrate accelerated setting within 1 minute upon placement within the prepared cavity in the mouth. The initial compressive strength shall be not less than 1000 psi at 5 minutes to withstand the condensation of dental amalgam.

The ultimate compressive strength attained shall be not less than 4000 psi. The volumetric change upon setting shall not exceed ± 5% and the pulp protector shall remain dimensionally stable and inert.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Mixing pad. The mixing pad shall be suitable for use in mixing pulp protector materials without absorption or contamination, and be of sufficient thickness to prevent tearing. The pad shall be sized to allow hand spatulation. The pad shall contain a minimum of 90 sheets of paper, which shall be bound on two surfaces with a suitable adhesive to ensure firm support during mixing, yet permit ease of removal of each sheet. In addition, each pad shall be mounted on a suitable cardboard.

The mixing and placement instrument. The mixing and placement instrument shall be fabricated of austenitic corrosion-resisting steel with the following dimensions:

overall length	4 to 4 1/2 inches
handle length	3 to 4 inches
handle diameter	1/6 to 1/4 inch
tip length	1/4 to 1/2 inch

The instrument tip shall be contra-angle, and shall taper to the distal end, and shall terminate in a small ball.

Finish. The handle shall be satin finished or fine ground finished to provide a finger grip, without knurling. Other portions of the metal shall be highly polished or bright smooth finished to facilitate cleaning.

Markings. Each tube of protector shall be permanently and legibly marked "Base" or "Catalyst".

Instructions. Detailed and accurate instructions shall accompany each package including storage instructions, method of application, and the effects of increased temperature and humidity on working and setting times.

Workmanship. The pulp protector shall be free of all contaminants, lumps or granules, or any other defect which detracts from its appearance or impairs its serviceability.

Unit. Package (PG). One package containing one tube of catalyst paste, one tube of base paste, a sufficient number of disposable mixing pad sheets, as specified, complete instructions for use and, when specified, one mixing and placement instrument, as specified, constitutes one unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for examination purposes shall be one package of pulp protector as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one package of pulp protector. Sampling shall be inspection level S-2 with an AQL of 1.0.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

NOTE: The following National Stock Numbers and their respective Item Identifications are covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION	TYPE
6520-00-878-2350	Pulp Protector, Dental, Calcium Hydroxide Paste; with instrument	I
6520-00-878-2351	Pulp Protector, Dental, Calcium Hydroxide Paste; without instrument	II

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6520-2543
Location: word\doc\paul\pulp

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54929	931217

3. DOCUMENT TITLE

PULP PROTECTOR, DENTAL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Defense Personnel Support Center ATTN: DPSC-MSE	(1) Commercial: (215)7372870 (2) AUTOVON: 444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.